Problem Set & Lab Report

Optimization Methods · Electromagnetic Waves

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Problem Set 3

Quadratic Minimization

Problem

Find the global minimum of

$$f(x) = x^2 - 4x + 5. \tag{1}$$

Solution. Set $f'(x) = 2x - 4 = 0$, so $x = 2$. Since $f''(x) = 2 > 0$, the critical point is a minimum, and $f(2) = 1$.

Problem

Show that $f(\boldsymbol{x}) = \boldsymbol{x}^T A \boldsymbol{x} - 2\boldsymbol{b}^T \boldsymbol{x}$ with A symmetric positive definite has a unique minimizer $\boldsymbol{x}^* = A^{-1}\boldsymbol{b}$.

Solution. The gradient is $\nabla f = 2A\boldsymbol{x} - 2\boldsymbol{b}$, so setting $\nabla f = 0$ gives $A\boldsymbol{x} = \boldsymbol{b}$. Since $A \succ 0$, the Hessian $\boldsymbol{H} = 2A$ is positive definite, hence $\boldsymbol{x}^* = A^{-1}\boldsymbol{b}$ is the unique minimizer.

Descent Directions

Definition Gradient Descent Update

The gradient descent iteration with step size α is

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha \nabla f(\boldsymbol{x}_k). \tag{1}$$

Problem

For $f(x) = x^2 - 4x + 5$ and $\alpha = 0.1$, starting from $x_0 = 0$, how many iterations does gradient descent take to reach $|f'(x)| < 10^{-3}$?

Solution. The update is $x_{k+1} = x_k - 0.2(x_k - 2) = 0.8x_k + 0.4$, converging to $x = 2$. The error $e_k = x_k - 2$ satisfies $e_{k+1} = 0.8 e_k$, so we need $0.8^k \cdot 2 < 10^{-3}$, i.e. $k > \frac{\ln(2000)}{\ln(\frac{1}{0.8})} \approx 34$. About 34 iterations.

Newton's Method

Remark

Newton's method uses the curvature of f to converge faster:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \boldsymbol{H}^{-1}_{f(\boldsymbol{x}_k)} \nabla f(\boldsymbol{x}_k). \tag{1}$$

Problem

Apply one Newton step to $f(x) = x^2 - 4x + 5$ from $x_0 = 0$ and compare with gradient descent.

Solution. $x_1 = x_0 - \frac{2x_0 - 4}{2} = 0 - \frac{-4}{2} = 2$. Newton's method reaches the exact minimizer in a single step for a quadratic function, while gradient descent needs 34 iterations.

Lab Report: Plane Wave Propagation

Setup and Governing Equation

Definition Plane Wave

In a lossless medium, the electric field of a plane wave propagating along z is

$$\boldsymbol{E}(z,t) = E_0 \cos(kz - \omega t)\hat{\boldsymbol{x}}. \tag{1}$$

Property Dispersion Relation

Substituting Eq (1) into the wave equation gives the dispersion relation $k = \frac{\omega}{c}$, $\quad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 2.998 \cdot 10^{8}\ \mathrm{m\,s^{-1}}$.

Theorem Energy Flux

The time-averaged Poynting vector of Eq (1) is

$$\angle(\boldsymbol{S})\angle = \frac{E_0^2}{2\mu_0 c}\hat{\boldsymbol{z}}, \tag{2}$$

representing the energy transported per unit area per unit time.

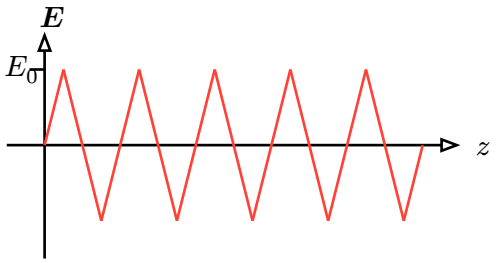
Experimental Measurement

Problem

A visible-light laser has a measured power of $5.0\,\mathrm{mW}$ over a spot of diameter $2.0\,\mathrm{mm}$. Estimate the peak electric field amplitude E_0.

Solution. Intensity is $I = \frac{P}{A} = \frac{5.0\cdot 10^{-3}}{\pi (1.0\cdot 10^{-3})^2} \approx 1592\,\mathrm{W\,m^{-2}}$. Using $I = \frac{E_0^2}{2\mu_0 c}$, $E_0 = \sqrt{2\mu_0 c I} \approx 1095\,\mathrm{V\,m^{-1}}$.

Numerical Verification



The snapshot above confirms the sinusoidal profile of Eq (1) at a fixed time $t = 0$.

Remark

For wavelengths in the visible range, $(380 - 750)\,\mathrm{nm}$, the corresponding photon energies are $(1.65 - 3.27)\,\mathrm{eV}$.